Cambridge (CIE) A Level Maths: Pure 1

Your notes

Differentiation

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- Definition of Gradient
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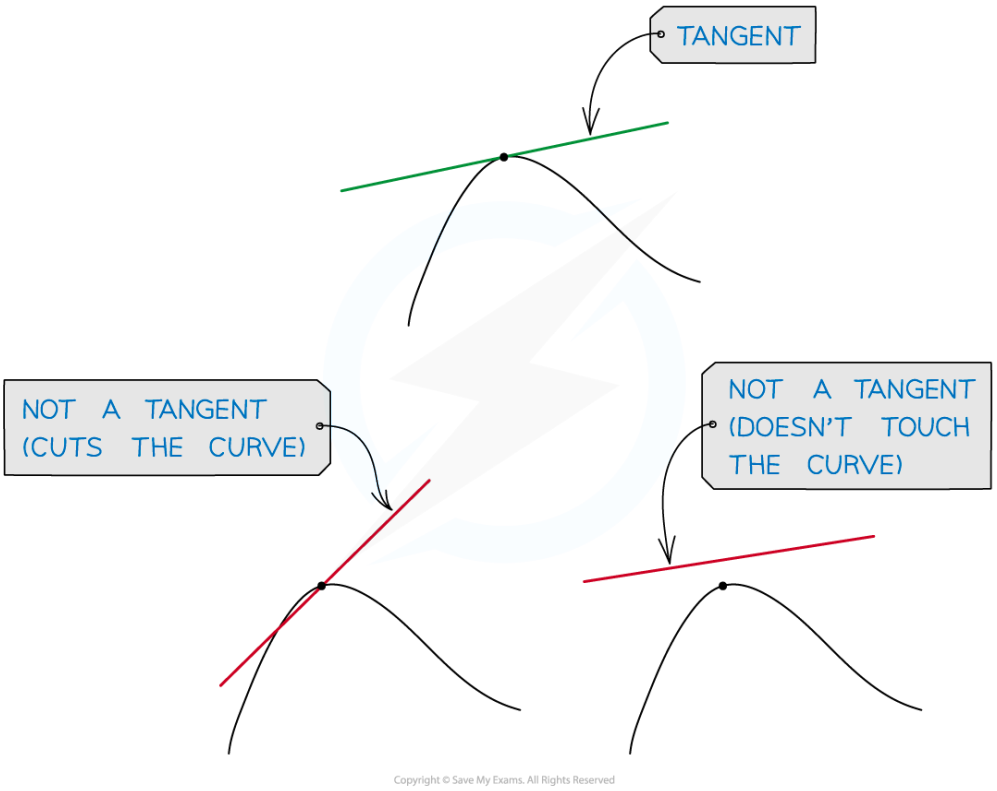
Your notes

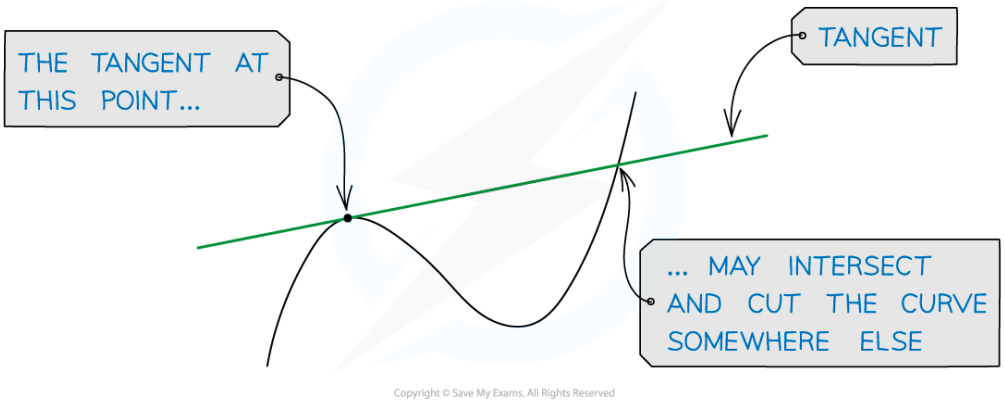
Definition of gradient

What is the gradient of a curve?

- At a given point the **gradient** of a curve is defined as the gradient of the **tangent** to the curve at that point
- A **tangent** to a curve is a line that just **touches** the curve at one point but doesn't cut the curve at that point



- A tangent *may* cut the curve somewhere else on the curve



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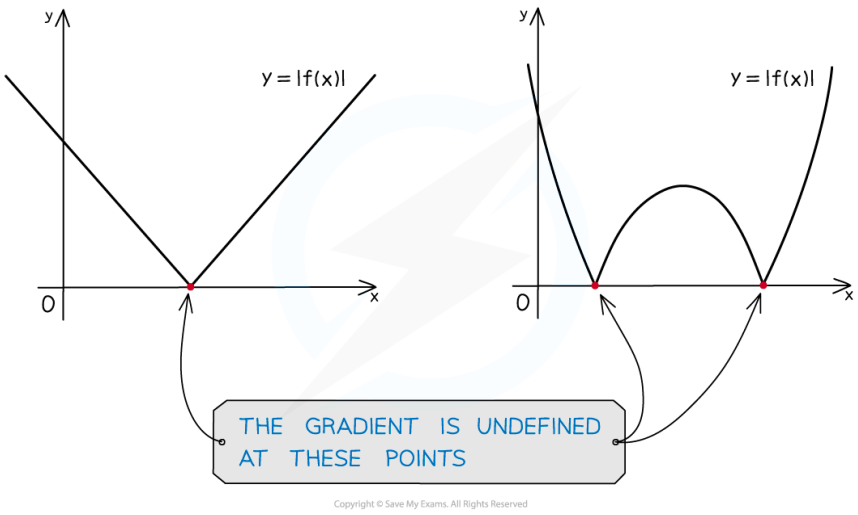
- It is only possible to draw **one** tangent to a curve at any given point
- Note that unlike the gradient of a straight line, the gradient of a curve is constantly changing

Examiner Tips and Tricks

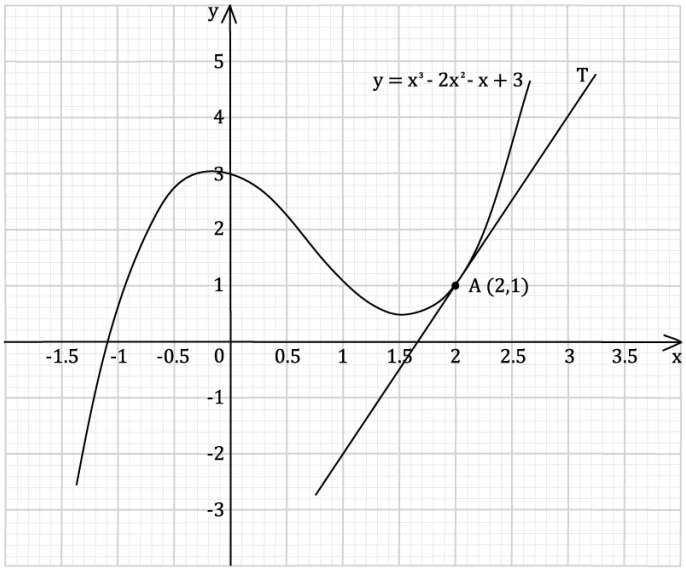
- A tangent only exists at points where a curve is smooth.
- For example, there is no tangent (or gradient) at one of the 'corners' in a modulus function.



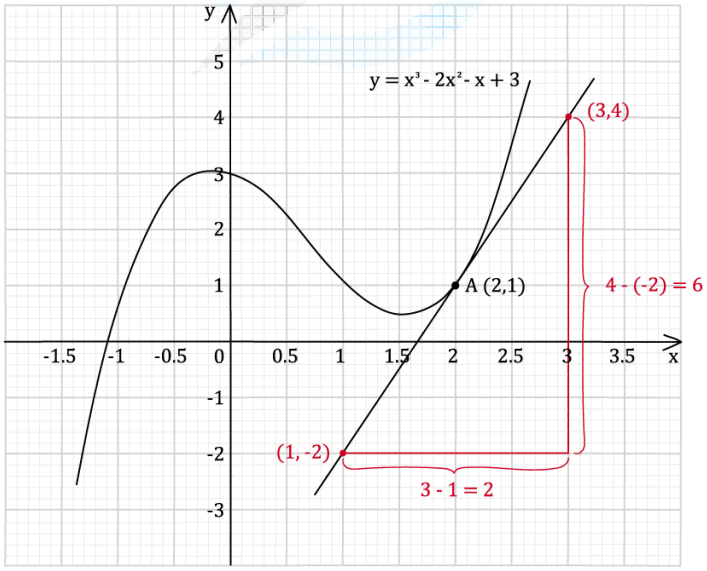
Worked Example

Your notes

The diagram shows the curve with equation $y = x^3 - 2x^2 - x + 3$. The tangent, T, to the curve at point $A(2,\ 1)$ is also shown.



Using the diagram, calculate the gradient of the curve at point A.



THE GRADIENT OF THE CURVE AT POINT A IS THE SAME AS THE GRADIENT OF THE TANGENT LINE.

SO

$$\text{GRADIENT} = \frac{\text{CHANGE IN } y}{\text{CHANGE IN } x} = \frac{4-(-2)}{3-1} = \frac{6}{2} = 3$$

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Definition of derivatives

What is the derivative of a function?

- **Differentiation** is an operation that calculates the **rate of change** of a **function** with respect to a variable
  - This means how much the function varies when the variable increases by one unit
- To **differentiate** a function (f(x)) with respect to the variable x we use the **notation**
  - $\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{f}(x))$
- The result is called the **derivative**

What is the link between derivatives and gradients?

- The rate of change of a function f(x) with respect to x can be thought of as the **gradient function** of the graph y = f(x)
  - We can write the gradient function (or derivative) as
    - $\mathrm{f}'(x)$ or $\frac{\mathrm{d}y}{\mathrm{d}x}$
- The rate of change of a function (or the gradient of its graph) varies for different values of x
  - For a **linear function** f(x) = mx + c the gradient is constant
    - We can write this as $\mathrm{f}'(x) = m$ or $\frac{\mathrm{d}y}{\mathrm{d}x} = m$
  - For the **quadratic function** f(x) = x^2 the gradient varies
    - Near the origin the gradient is close to 0
    - As x increases the gradient of the graph increases

How can I find the derivative of a function at a point?

- The **derivative** of a function (or gradient of its graph) at a point is equal to the **gradient of the tangent** to the graph at that point
- To **estimate** the gradient you could **draw the tangent** and calculate its **gradient**
- To find the **actual gradient** at a point x
  - Pick a **second point** on the curve close to the first point (call it $x + h$)
  - Calculate the **gradient of the chord** joining the two points

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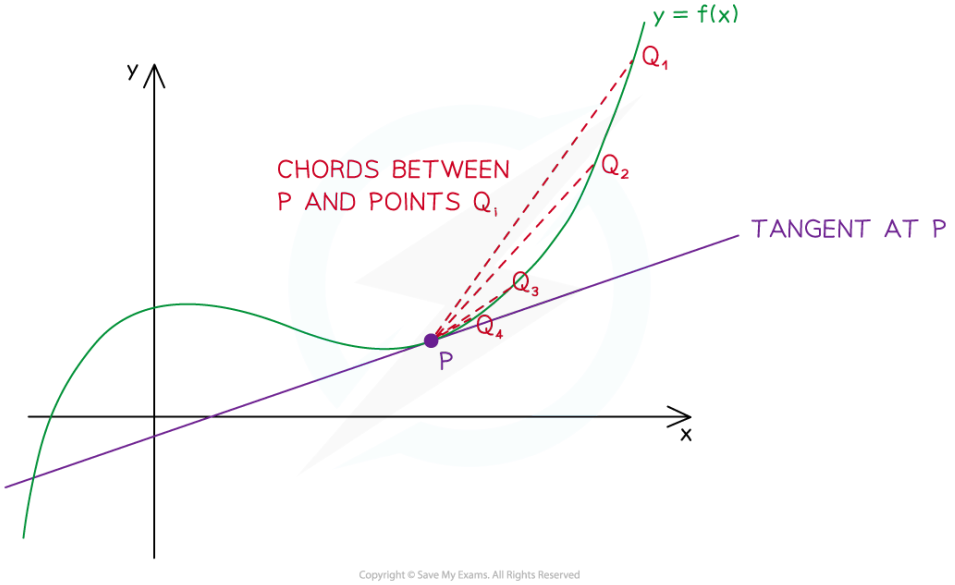
- $\dfrac{f(x+h)-f(x)}{(x+h)-x} = \dfrac{f(x+h)-f(x)}{h}$
- Move the **second point closer to the first point** (make h get close to zero)
- Examine what **happens to the gradient** of the chord
- The gradient of the tangent will be the **limit of the gradients of the chords**
  - $f'(x) = \lim_{h \to 0} \dfrac{f(x+h)-f(x)}{h}$
    - You do **not** need to remember this formula



Your notes

Worked Example

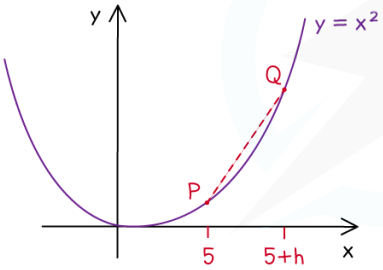
Your notes

The curve C is given by the equation $y = x^2$. The points P and Q lie on C and have x coordinates 5 and $(5 + h)$ respectively, where $h > 0$.

(a) Find, in terms of h, the gradient of the chord joining P and Q.

(b) Hence explain why the gradient of the tangent to C at P is 10.

a) QUICK SKETCH



FIND THE y–COORDINATES

$x = 5$: $y = 5^2 = 25$

$x = 5 + h$: $y = (5 + h)^2 = 25 + 10h + h^2$

FIND THE GRADIENT BETWEEN P AND Q

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(\cancel{25} + 10h + h^2) - \cancel{25}}{(\cancel{5} + h) - \cancel{5}} = \frac{10h + h^2}{h} = \frac{\cancel{h}(10+h)}{\cancel{h}}$$

GRADIENT $= 10 + h$

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b) THE GRADIENT OF THE TANGENT AT P IS EQUAL TO THE LIMIT OF THE GRADIENT OF THE CHORD JOINING P AND Q AS Q GETS CLOSER TO P.

AS Q GETS CLOSER TO P, $(5 + h)$ GETS CLOSER TO 5 SO h GETS CLOSER TO 0.

AS h GETS CLOSE TO 0, $10 + h$ LIMITS 10.

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Examiner Tips and Tricks

- Deriving a derivative from scratch is **not examinable**
- This revision note is intended to give you an understanding of what derivatives do

Your notes

Differentiating powers of x

How do I differentiate expressions with powers of x?

- Powers of ***x*** are differentiated according to the following formulae:

IF n IS ANY REAL NUMBER, AND a IS A CONSTANT, THEN

1. FOR $f(x) = x^n$, $f'(x) = nx^{n-1}$
   FOR $y = x^n$, $\frac{dy}{dx} = nx^{n-1}$
2. FOR $f(x) = ax^n$, $f'(x) = anx^{n-1}$
   FOR $y = ax^n$, $\frac{dy}{dx} = anx^{n-1}$

> $f(x)$ AND $\frac{dy}{dx}$ ARE BOTH WAYS OF WRITING THE DERIVATIVE, AND THE PAIRS OF FORMULAE UNDER EACH NUMBER ABOVE ARE ENTIRELY EQUIVALENT. WE TEND TO TALK ABOUT $f(x)$ WHEN WE HAVE OUR "FUNCTIONS" HAT ON, AND ABOUT $y=...$ WHEN WE HAVE OUR "GRAPHS" HAT ON – BUT ULTIMATELY THEY ARE JUST DIFFERENT NOTATIONS FOR THE SAME THING.

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- Differentiating terms with positive integer powers of ***x*** is straightforward

$f(x) = x^2 \longrightarrow f'(x) = 2x^{2-1} = 2x$

$f(x) = 5x^3 \longrightarrow f'(x) = 5 \times 3x^{3-1} = 15x^2$

$y = -4x^5 \longrightarrow \frac{dy}{dx} = -4 \times 5x^{5-1} = -20x^4$

$y = \frac{2}{3}x^7 \longrightarrow \frac{dy}{dx} = \frac{2}{3} \times 7x^{7-1} = \frac{14}{3}x^6$

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- But negative and fractional powers of ***x*** can also be differentiated this way

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$$f(x) = 2x^{-1} \longrightarrow f'(x) = 2 \times (-1)x^{-1-1} = -2x^{-2}$$

$$f(x) = 5\sqrt{x} = 5x^{\frac{1}{2}} \longrightarrow f'(x) = 5 \times \frac{1}{2}x^{\frac{1}{2}-1} = \frac{5}{2}x^{-\frac{1}{2}} = \frac{5}{2\sqrt{x}}$$

$$\sqrt{x} = x^{\frac{1}{2}}$$

$$y = \frac{4}{5}x^{\frac{2}{3}} \longrightarrow \frac{dy}{dx} = \frac{4}{5} \times \frac{2}{3}x^{\frac{2}{3}-1} = \frac{8}{15}x^{-\frac{1}{3}}$$

$$y = \frac{1}{x^4} = x^{-4} \longrightarrow \frac{dy}{dx} = -4x^{-4-1} = -4x^{-5} = -\frac{4}{x^5}$$

$$\frac{1}{x^p} = x^{-p}$$

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- Don't forget these two special cases:

IF a IS CONSTANT, THEN

3. FOR $f(x) = ax$, $f'(x) = a$

   FOR $y = ax$, $\frac{dy}{dx} = a$

4. FOR $f(x) = a$, $f'(x) = 0$

   FOR $y = a$, $\frac{dy}{dx} = 0$

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- These allow you to differentiate constants and linear terms in x

$$f(x) = 3 \longrightarrow f'(x) = 0$$

$$f(x) = -5x \longrightarrow f'(x) = -5$$

$$y = \sqrt{7} \longrightarrow \frac{dy}{dx} = 0$$

$$y = x \longrightarrow \frac{dy}{dx} = 1$$

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- If the expression involves a sum or difference of terms just differentiate one term at a time

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$$f(x) = 3x^2 - 5x + 3$$

$$f'(x) = 6x - 5$$

THE DERIVATIVE OF 3 (A CONSTANT) IS ZERO

$$y = x^4 - \sqrt{x} + \frac{4}{x}$$

$$\frac{dy}{dx} = 4x^3 - \frac{1}{2\sqrt{x}} - \frac{4}{x^2}$$

IN GENERAL IF $y = f(x) + g(x) + h(x) + \ldots$

THEN $\frac{dy}{dx} = f'(x) + g'(x) + h'(x) + \ldots$

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Examiner Tips and Tricks

- Take extra care when differentiating negative and fractional powers of ***x*** as mishandling negative signs and fractions is a common way to lose marks in these sorts of questions.

Worked Example

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$f(x) = \dfrac{8}{q\sqrt{x}} + \sqrt{x}$, where q is a real constant and $x > 0$.

Given that $f'(3) = -\dfrac{\sqrt{3}}{18}$, find the value of q.

$$f(x) = \frac{8}{q}x^{-\frac{1}{2}} + x^{\frac{1}{2}}$$

$\sqrt{x} = x^{\frac{1}{2}}$ $\quad \dfrac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$

$$f'(x) = -\frac{1}{2} \times \frac{8}{q}x^{-\frac{3}{2}} + \frac{1}{2} \times x^{-\frac{1}{2}}$$

$$= \frac{1}{2}x^{-\frac{1}{2}} - \frac{4}{q}x^{-\frac{3}{2}}$$

$$= \frac{1}{2\sqrt{x}} - \frac{4}{qx\sqrt{x}}$$

$x^{\frac{3}{2}} = x\sqrt{x}$

$$f'(3) = \frac{1}{2\sqrt{3}} - \frac{4}{3q\sqrt{3}} = \frac{3q-8}{6q\sqrt{3}}$$

REWRITE AS SINGLE FRACTION WITH COMMON DENOMINATOR

SO $\dfrac{3q-8}{6q\sqrt{3}} = -\dfrac{\sqrt{3}}{18}$

$$3q-8 = -\frac{\sqrt{3}}{18}\left(6q\sqrt{3}\right) = -q$$

$$4q = 8$$

$$q = 2$$

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Chain Rule

What is the chain rule?

- If ***y*** is a function of ***u***, and ***u*** is a function of ***x***, then the **chain rule** tells us that

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

- The chain rule allows us to differentiate more complicated expressions and **composite functions**
- You will often see and use the **chain rule** with different variables
  - This is particularly useful for **connected rates of change**

How do I differentiate $(ax + b)^n$?

- For $n = 2$ you will most likely **expand the brackets** and differentiate each term separately
- If $n > 2$ this becomes **time-consuming** and if n is not a positive integer we need a different method completely
- The chain rule allows us to use **substitution** to differentiate any function in the form $y = (ax + b)^n$
  - Let $u = ax + b$, then $y = u^n$
  - **Differentiate** both parts **separately**
    - $\frac{du}{dx} = a$ and $\frac{dy}{du} = nu^{n-1}$
  - Put both parts into the **chain rule**
    - $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = a \times nu^{n-1} = anu^{n-1}$
  - **Substitute** $u = ax + b$ back into your answer
    - $\frac{dy}{dx} = an(ax + b)^{n-1}$

How do I differentiate √(ax+b)?

- The chain rule allows us to use **substitution** to differentiate any function in the form $y = \sqrt{ax + b}$
- Rewrite $\sqrt{ax + b} = (ax + b)^{\frac{1}{2}}$

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- Let $u = ax + b$, then $y = u^{½}$
- **Differentiate** both parts **separately**
  - $\frac{du}{dx} = a$ and $\frac{dy}{du} = \frac{1}{2}u^{-\frac{1}{2}}$
- Put both parts into the **chain rule**
  - $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = a \times \frac{1}{2}u^{-\frac{1}{2}} = \frac{a}{2}u^{-\frac{1}{2}}$
- **Substitute** $u = ax + b$ back into your answer
  - $\frac{dy}{dx} = \frac{a}{2}(ax+b)^{-\frac{1}{2}} = \frac{a}{2\sqrt{ax+b}}$

- This method can be used for any **fractional power** of any linear or non-linear expression
  - Provided you know how to differentiate the non-linear expression

How do I differentiate $(f(x))^n$?

- This method can be used for any linear or non – linear expression
  - Let $u = f(x)$ and follow the method above
  - In general if $y = (f(x))^n$ then $\frac{dy}{dx} = nf'(x)(f(x))^{n-1}$
- With practice you will be able to carry out this method **without the need for *u***
  - This is essential for learning the **reverse chain rule** later in the course

Worked Example

Your notes

Your notes

Find the gradient of the curve $y = \frac{1}{\sqrt{2x^2 - 4}}$ at the point (2, 0.5)

$$y = \frac{1}{\sqrt{2x^2 - 4}} = (2x^2 - 4)^{-\frac{1}{2}}$$

REWRITE IN THE FORM $y = (fx)^n$

METHOD 1: USING SUBSTITUTION

LET $u = 2x^2 - 4$ THEN $y = u^{-\frac{1}{2}}$

$\frac{du}{dx} = 4x$ $\frac{dy}{du} = -\frac{1}{2}u^{-\frac{3}{2}}$

USE THE CHAIN RULE TO FIND $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du} = 4x \times -\frac{1}{2}u^{-\frac{3}{2}}$$

$$= -2x(u)^{-\frac{3}{2}}$$

SUBSTITUTE $u = 2x^2 - 4$ BACK IN

$$\frac{dy}{dx} = -2x(2x^2 - 4)^{-\frac{3}{2}}$$

$$= \frac{-2x}{\sqrt{(2x^2 - 4)^3}}$$

SOMETIMES IT IS EASIER TO LEAVE IN THIS FORM, DEPENDING ON THE NEXT PART OF THE QUESTION

SUBSTITUTE $x = 2$ INTO $\frac{dy}{dx}$

$$\frac{-2(2)}{\sqrt{(2(2)^2 - 4)^3}} = -\frac{4}{8} = -0.5$$

$\frac{dy}{dx} = -0.5$ AT THE POINT (2, 0.5)

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METHOD 2: WITHOUT SUBSTITUTION:

$$y = (2x^2 - 4)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = -\frac{1}{2}(2x^2 - 4)^{-\frac{3}{2}} \times 4x$$

DIFFERENTIATE 'MAIN FUNCTION'

DERIVATIVE OF THE INSIDE.

$$= -2x(2x^2 - 4)^{-\frac{3}{2}}$$

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Examiner Tips and Tricks

If using *u* as a substitution don't forget to substitute *x* back into your final answer.